OUTPUTS

Google Colab Outputs:

```
Saving disease_dataset.csv to disease_dataset.csv
```

```python
df = pd.read_csv("disease_dataset.csv")
df.head()
```

	Patient_ID	Age	Gender	Fever	Cough	Headache	BP	Sugar	Cholesterol	Medical_History	Disease
0	1	25	Male	1	1	0	120	85	180	NaN	Flu
1	2	45	Female	0	1	1	140	160	220	Diabetes	Heart Disease
2	3	35	Male	1	1	1	130	120	200	NaN	Flu
3	4	50	Female	0	0	1	150	180	240	Hypertension	Heart Disease
4	5	23	Male	1	1	0	110	90	170	NaN	Flu

```python
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 11 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Patient_ID       20 non-null     int64
 1   Age              20 non-null     int64
 2   Gender           20 non-null     object
 3   Fever            20 non-null     int64
 4   Cough            20 non-null     int64
```

Terminal

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 11 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Patient_ID       20 non-null     int64
 1   Age              20 non-null     int64
 2   Gender           20 non-null     object
 3   Fever            20 non-null     int64
 4   Cough            20 non-null     int64
 5   Headache         20 non-null     int64
 6   BP               20 non-null     int64
 7   Sugar            20 non-null     int64
 8   Cholesterol      20 non-null     int64
 9   Medical_History  10 non-null     object
 10  Disease          20 non-null     object
dtypes: int64(8), object(3)
memory usage: 1.8+ KB
None
       Patient_ID        Age      Fever      Cough   Headache          BP  \
count    20.00000  20.000000  20.000000  20.000000  20.000000   20.000000
mean     10.50000  40.350000   0.500000   0.700000   0.700000  136.500000
std       5.91608  11.636037   0.512989   0.470162   0.470162   13.492688
min       1.00000  23.000000   0.000000   0.000000   0.000000  110.000000
25%       5.75000  30.500000   0.000000   0.000000   0.000000  127.250000
50%      10.50000  39.500000   0.500000   1.000000   1.000000  136.500000
75%      15.25000  49.250000   1.000000   1.000000   1.000000  146.500000
max      20.00000  60.000000   1.000000   1.000000   1.000000  160.000000

            Sugar  Cholesterol
count   20.000000    20.000000
mean   140.250000   213.250000
```

Variables Terminal

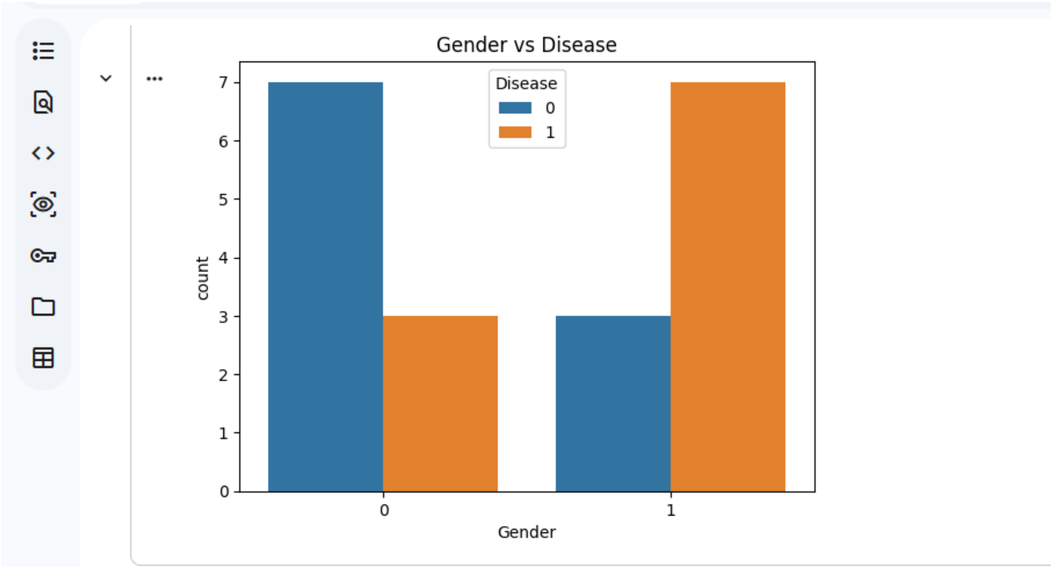
```python
df["Disease"] = le.fit_transform(df["Disease"])

df.head()
```

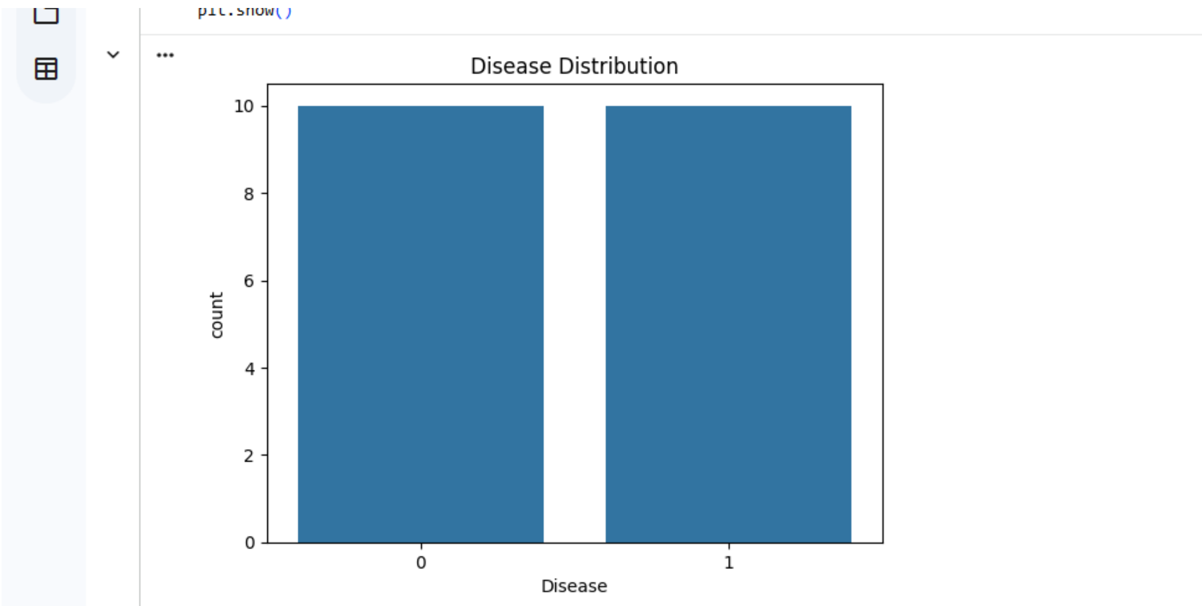
```
Patient_ID          0
Age                 0
Gender              0
Fever               0
Cough               0
Headache            0
BP                  0
Sugar               0
Cholesterol         0
Medical_History    10
Disease             0
dtype: int64
```

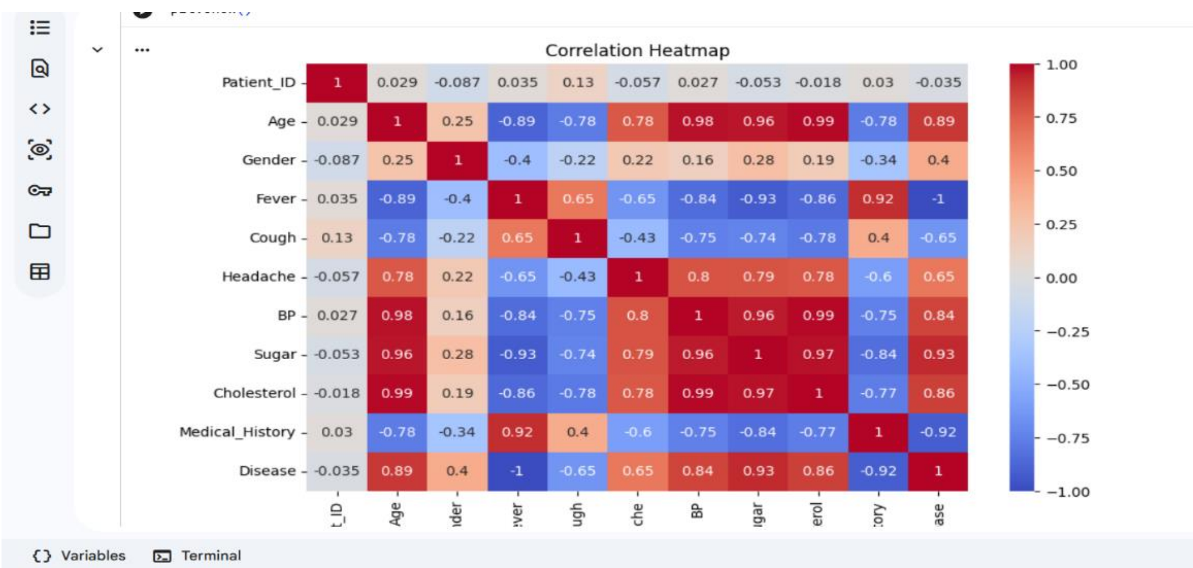
	Patient_ID	Age	Gender	Fever	Cough	Headache	BP	Sugar	Cholesterol	Medical_History	Disease
0	1	25	1	1	1	0	120	85	180	2	0
1	2	45	0	0	1	1	140	160	220	0	1
2	3	35	1	1	1	1	130	120	200	2	0
3	4	50	0	0	0	1	150	180	240	1	1
4	5	23	1	1	1	0	110	90	170	2	0

```python
# Disease distribution
sns.countplot(x='Disease', data=df)
plt.title("Disease Distribution")
```

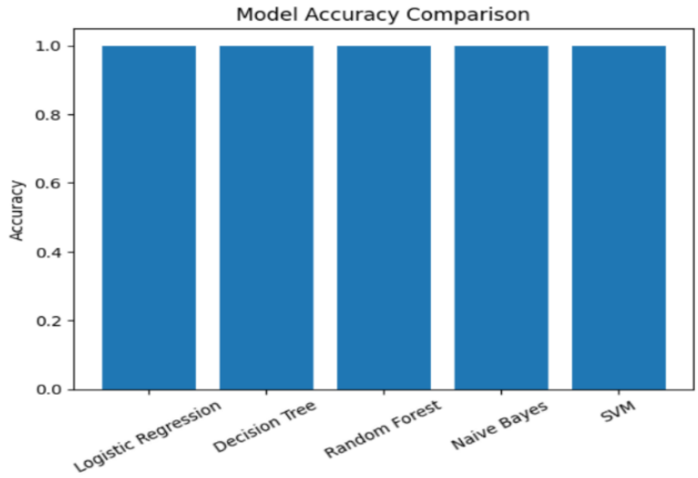
Variables Terminal


Gender vs Disease
Disease
0
1
count
Gender


plt.show()
Disease Distribution
count
Disease

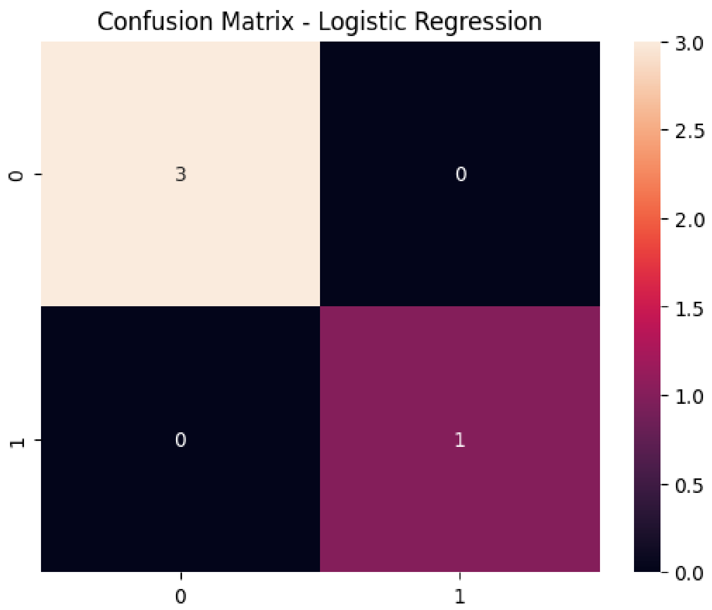

Correlation Heatmap
Patient_ID
Age
Gender
Fever
Cough
Headache
BP
Sugar
Cholesterol
Medical_History
Disease
Variables
Terminal

```
plt.ylabel("Accuracy")
plt.show()
```

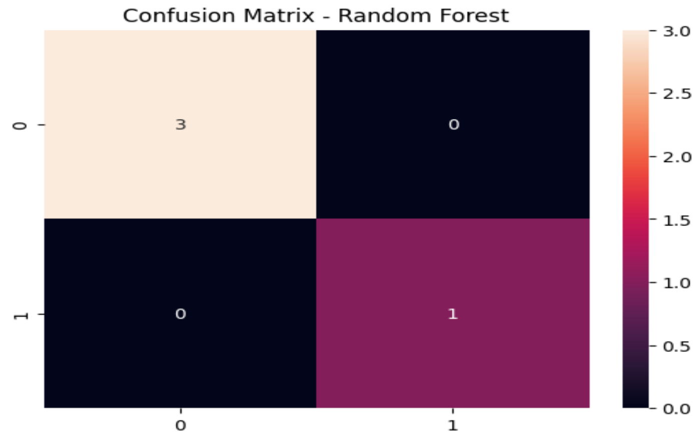


Variables Terminal

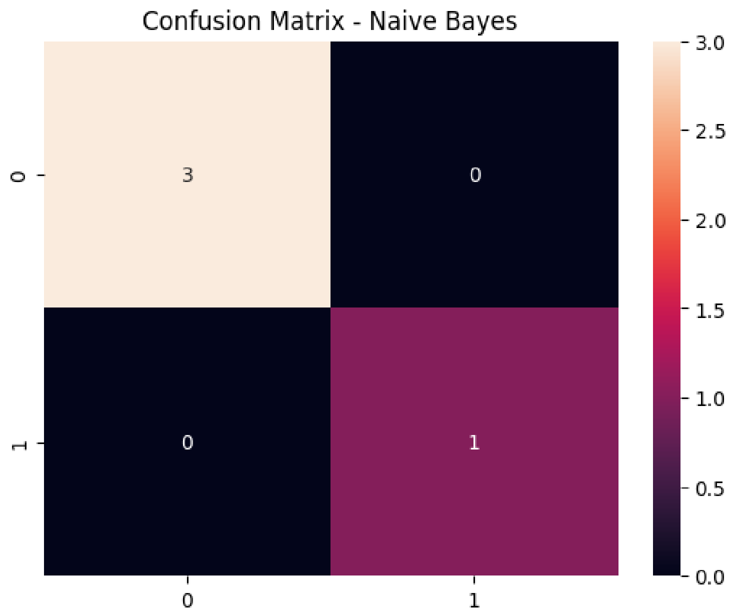
```
Model: Logistic Regression
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         1

    accuracy                           1.00         4
   macro avg       1.00      1.00      1.00         4
weighted avg       1.00      1.00      1.00         4
```



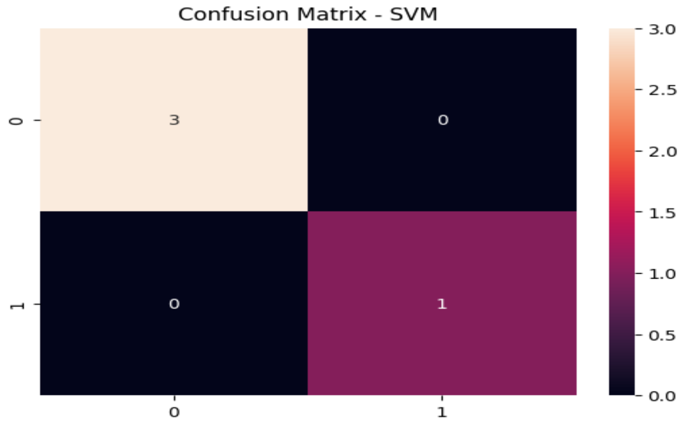
```
Model: Random Forest
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         1

    accuracy                           1.00         4
   macro avg       1.00      1.00      1.00         4
weighted avg       1.00      1.00      1.00         4
```

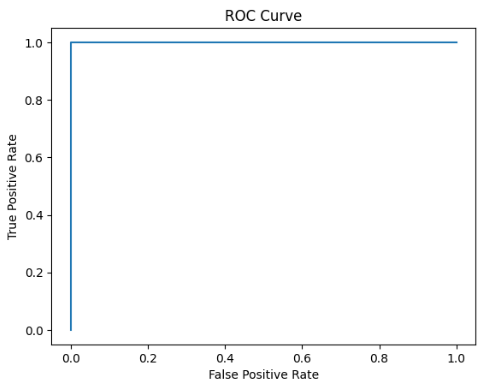



```
Model: Naive Bayes
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         1

    accuracy                           1.00         4
   macro avg       1.00      1.00      1.00         4
weighted avg       1.00      1.00      1.00         4
```



```
Model: SVM
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         1

    accuracy                           1.00         4
   macro avg       1.00      1.00      1.00         4
weighted avg       1.00      1.00      1.00         4
```





Streamlit Web Outputs:

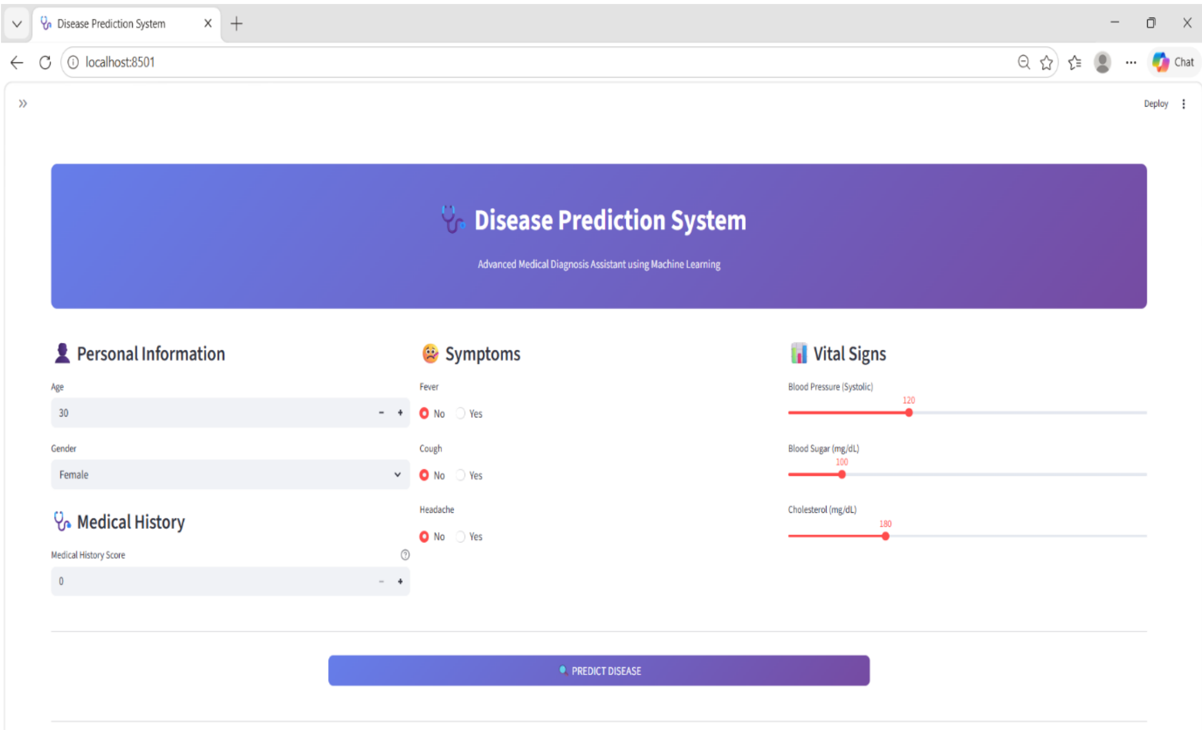
Commands to run on command prompt;

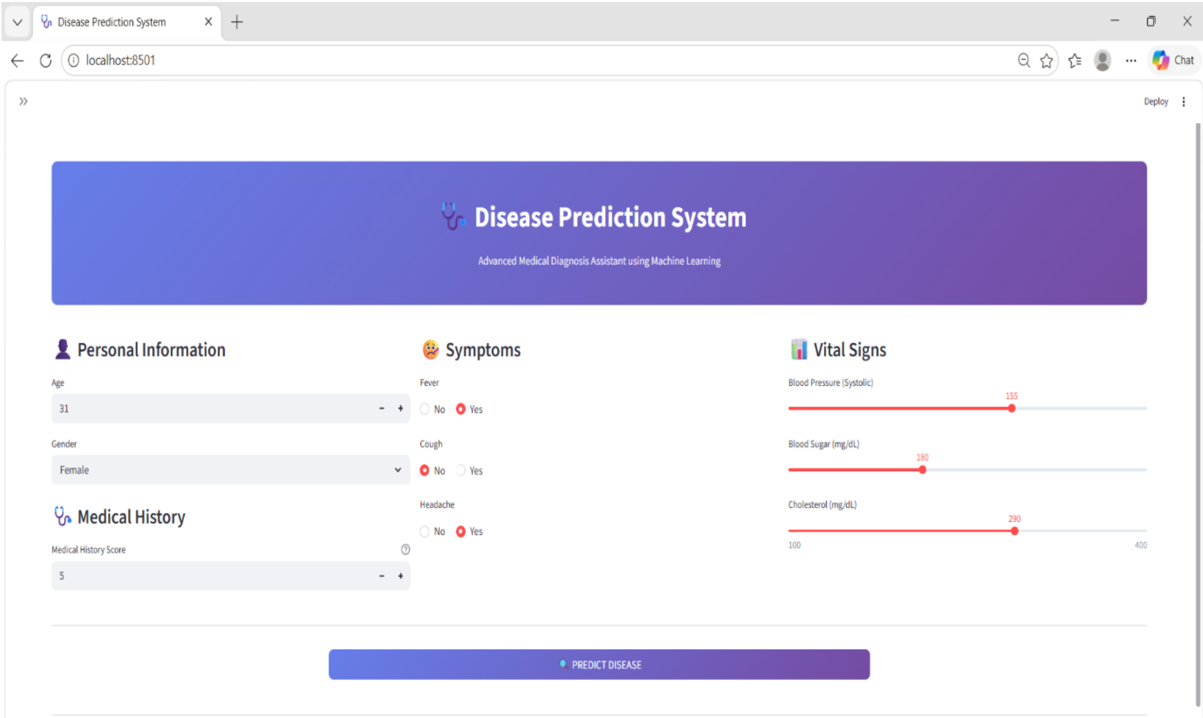
cd D:

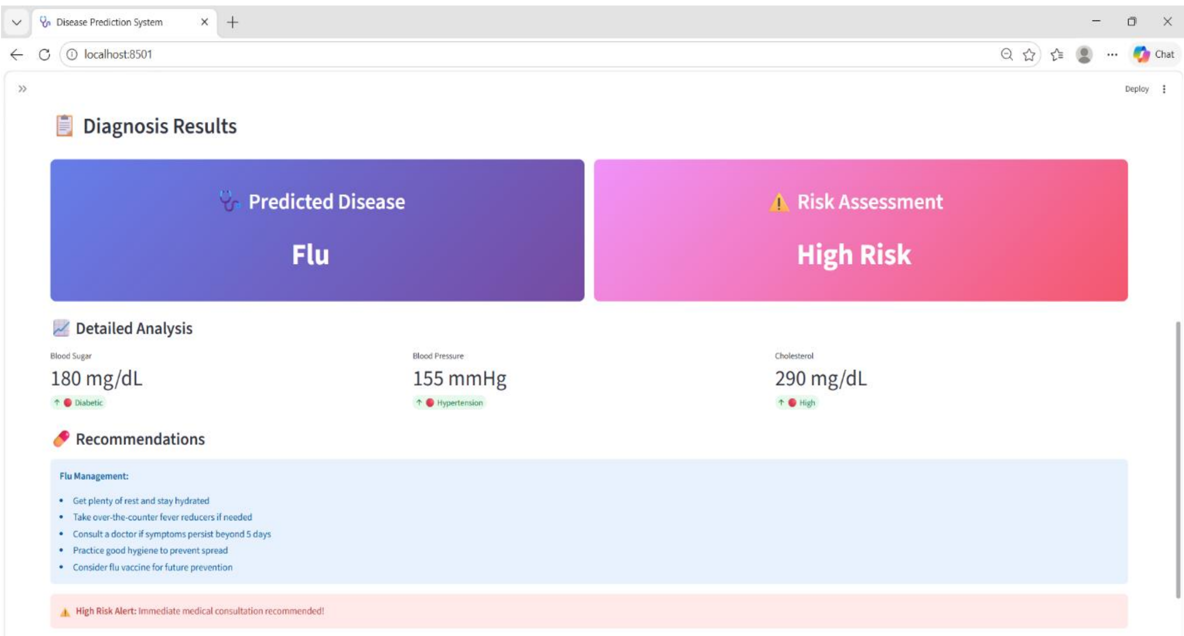
D:

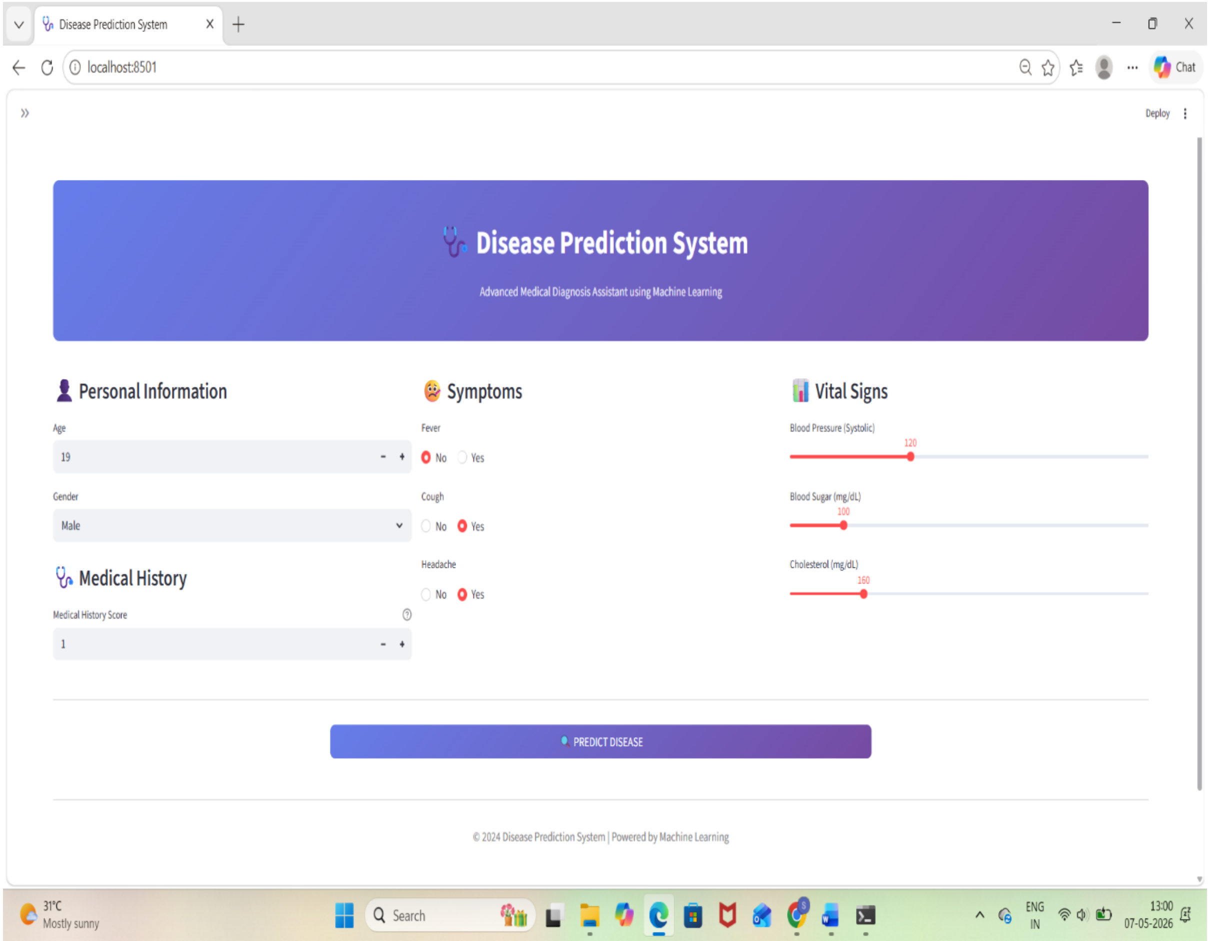
cd D:\disease predection

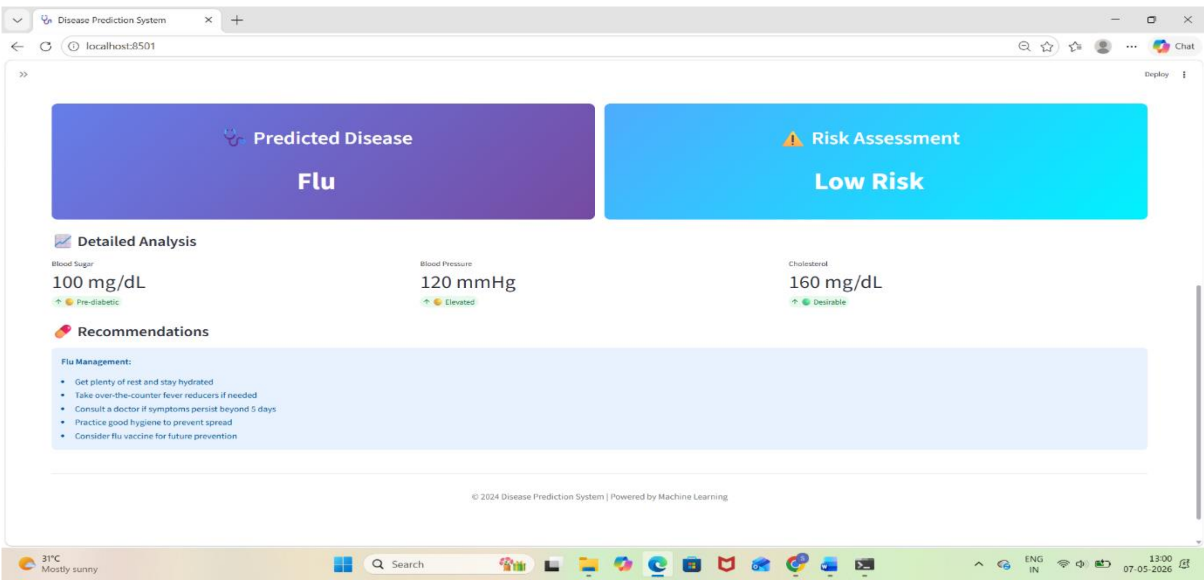
python -m streamlit run app.py

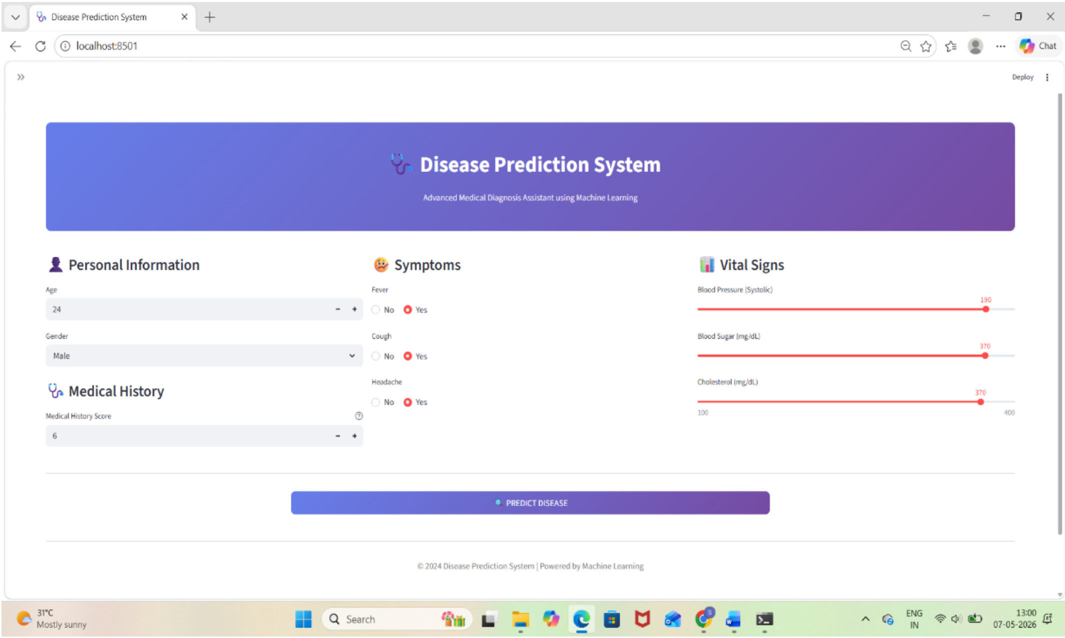
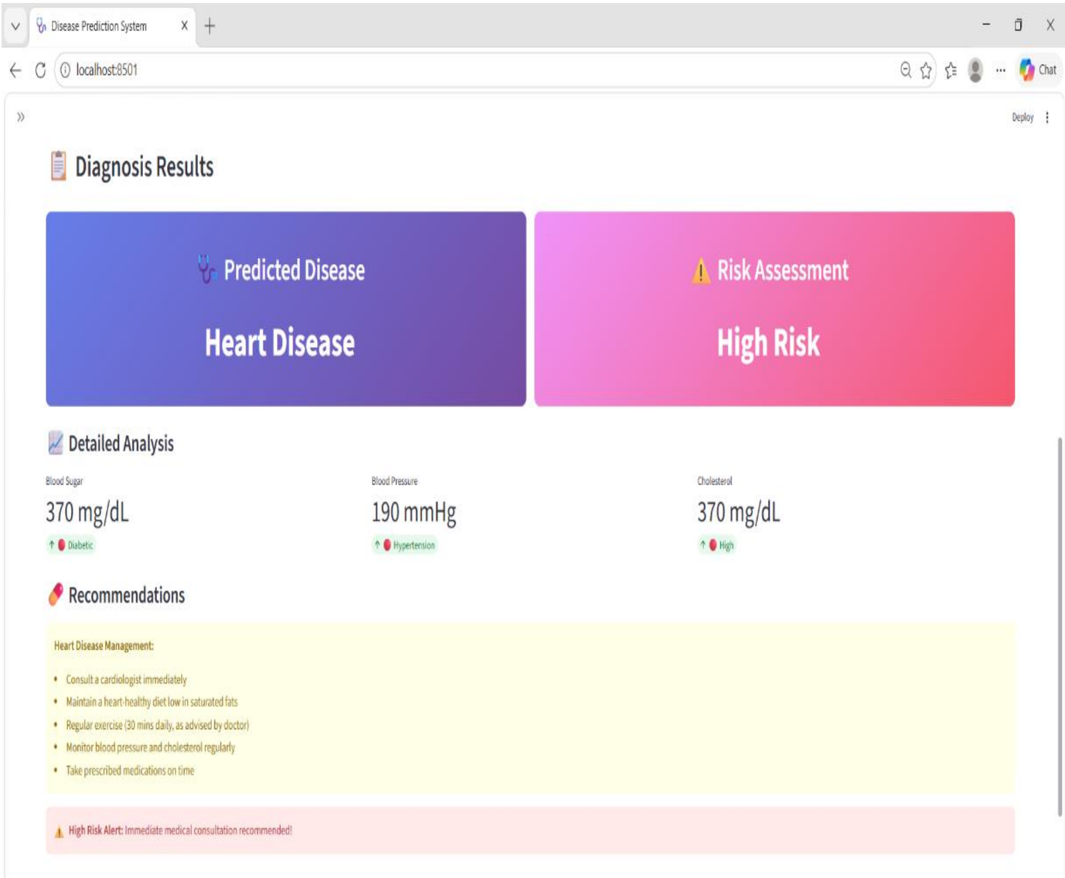


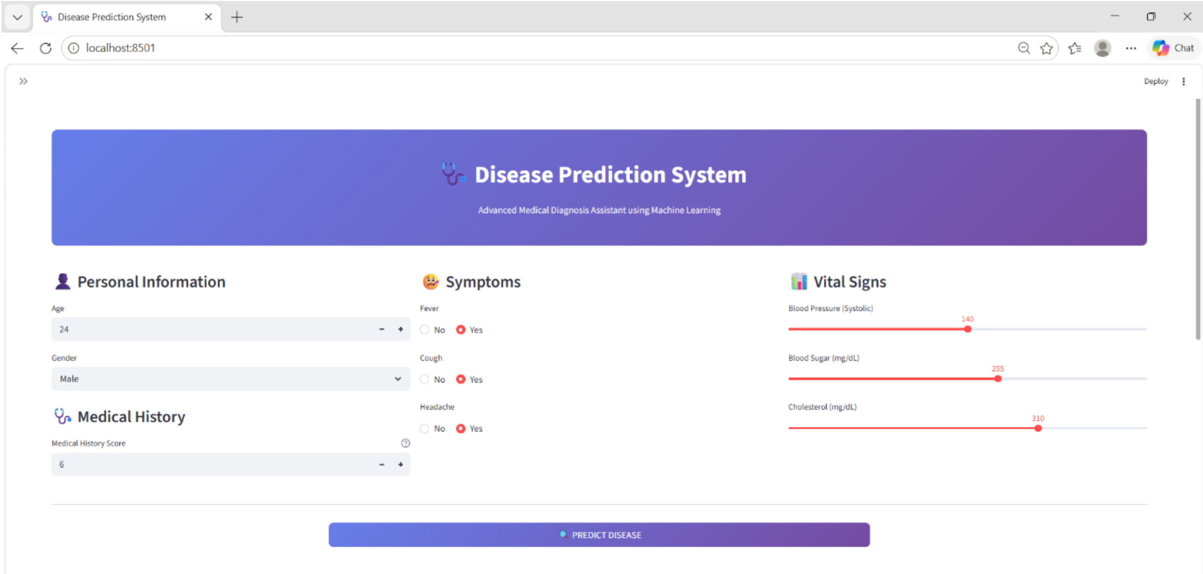

Disease Prediction System
localhost:8501
Deploy
Disease Prediction System
Advanced Medical Diagnosis Assistant using Machine Learning
Personal Information
Age
31
Gender
Female
Medical History
Medical History Score
5
Symptoms
Fever
No
Yes
Cough
No
Yes
Headache
No
Yes
Vital Signs
Blood Pressure (Systolic)
155
Blood Sugar (mg/dL)
180
Cholesterol (mg/dL)
290
100
400
PREDICT DISEASE

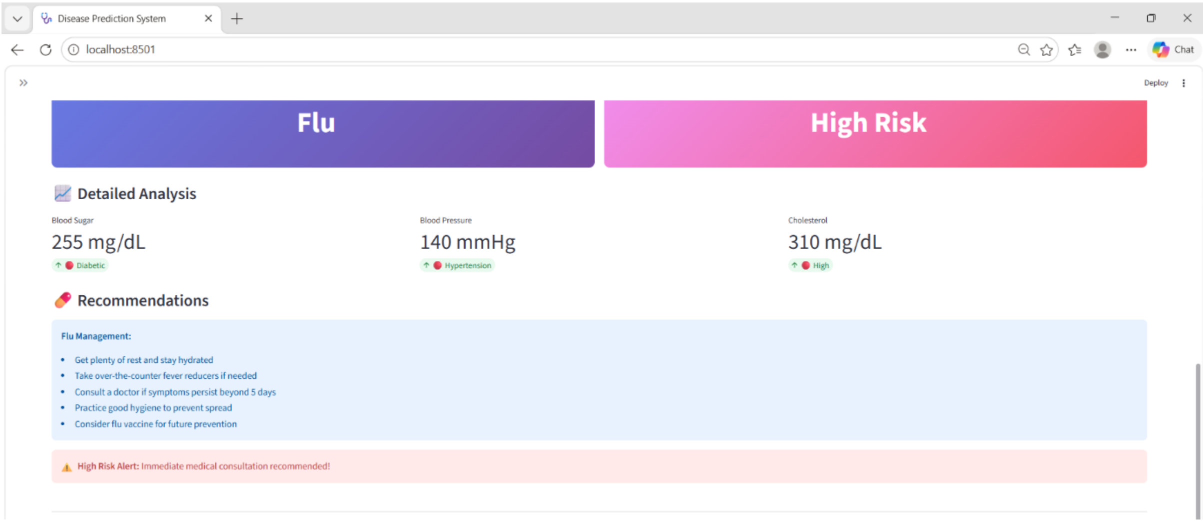

Disease Prediction System
localhost:8501
Deploy
Diagnosis Results
Predicted Disease
Flu
Risk Assessment
High Risk
Detailed Analysis
Blood Sugar
180 mg/dL
Diabetic
Blood Pressure
155 mmHg
Hypertension
Cholesterol
290 mg/dL
High
Recommendations
Flu Management:
Get plenty of rest and stay hydrated
Take over-the-counter fever reducers if needed
Consult a doctor if symptoms persist beyond 5 days
Practice good hygiene to prevent spread
Consider flu vaccine for future prevention
High Risk Alert: Immediate medical consultation recommended!


Disease Prediction System
localhost:8501
Deploy
Disease Prediction System
Advanced Medical Diagnosis Assistant using Machine Learning
Personal Information
Age
19
Gender
Male
Medical History
Medical History Score
1
Symptoms
Fever
No
Yes
Cough
No
Yes
Headache
No
Yes
Vital Signs
Blood Pressure (Systolic)
120
Blood Sugar (mg/dL)
100
Cholesterol (mg/dL)
160
PREDICT DISEASE
© 2024 Disease Prediction System | Powered by Machine Learning
31°C
Mostly sunny
Search
ENG
IN
13:00
07-05-2026


Disease Prediction System
localhost:8501
Deploy
Predicted Disease
Flu
Risk Assessment
Low Risk
Detailed Analysis
Blood Sugar
100 mg/dL
Pre-diabetic
Blood Pressure
120 mmHg
Elevated
Cholesterol
160 mg/dL
Desirable
Recommendations
Flu Management:
Get plenty of rest and stay hydrated
Take over-the-counter fever reducers if needed
Consult a doctor if symptoms persist beyond 5 days
Practice good hygiene to prevent spread
Consider flu vaccine for future prevention
© 2024 Disease Prediction System | Powered by Machine Learning
31°C
Mostly sunny
Search
ENG
IN
13:00
07-05-2026


Disease Prediction System
localhost:8501
Deploy
Disease Prediction System
Advanced Medical Diagnosis Assistant using Machine Learning
Personal Information
Age
24
Gender
Male
Medical History
Medical History Score
6
Symptoms
Fever
No
Yes
Cough
No
Yes
Headache
No
Yes
Vital Signs
Blood Pressure (Systolic)
190
Blood Sugar (mg/dL)
370
Cholesterol (mg/dL)
370
100
400
PREDICT DISEASE
© 2024 Disease Prediction System | Powered by Machine Learning
31°C
Mostly sunny
Search
ENG
IN
13:00
07-05-2026

Disease Prediction System
localhost:8501
Chat
Deploy
Diagnosis Results
Predicted Disease
Heart Disease
Risk Assessment
High Risk
Detailed Analysis
Blood Sugar
370 mg/dL
Diabetic
Blood Pressure
190 mmHg
Hypertension
Cholesterol
370 mg/dL
High
Recommendations
Heart Disease Management:
Consult a cardiologist immediately
Maintain a heart-healthy diet low in saturated fats
Regular exercise (30 mins daily, as advised by doctor)
Monitor blood pressure and cholesterol regularly
Take prescribed medications on time
High Risk Alert: Immediate medical consultation recommended!


Disease Prediction System
localhost:8501
Deploy
Disease Prediction System
Advanced Medical Diagnosis Assistant using Machine Learning
Personal Information
Age
24
Gender
Male
Medical History
Medical History Score
6
Symptoms
Fever
No
Yes
Cough
No
Yes
Headache
No
Yes
Vital Signs
Blood Pressure (Systolic)
140
Blood Sugar (mg/dL)
255
Cholesterol (mg/dL)
310
PREDICT DISEASE


Disease Prediction System
localhost:8501
Deploy
Flu
High Risk
Detailed Analysis
Blood Sugar
255 mg/dL
Diabetic
Blood Pressure
140 mmHg
Hypertension
Cholesterol
310 mg/dL
High
Recommendations
Flu Management:
Get plenty of rest and stay hydrated
Take over-the-counter fever reducers if needed
Consult a doctor if symptoms persist beyond 5 days
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High Risk Alert: Immediate medical consultation recommended!